Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer and USC MSCS graduate with 2+ years building **Python, Java, and TypeScript** applications, cloud services, SQL data pipelines, automated tests, and secure digital experiences.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Data Structures, Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Web Technologies, Computer Networks, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built an Azure-backed digital platform with normalized schemas, row-level isolation, and indexed **SQL data models**, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ daily users.
- Reworked asynchronous Python services with FastAPI workers, explicit job states, retries, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **cloud application reliability** in production.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed React and Flask workflows for uploads, history, model selection, and feedback, supporting 150+ users and 10K+ daily requests through secure **full-stack application features** and REST APIs.
- Designed concurrent Python pipelines for secure file ingestion and multimodal processing, handling 1K+ files per batch with provider fallbacks, validation, and **automated integration testing** across production workflows.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built reproducible Python pipelines processing 100K+ autonomous-driving logs, improving planning 18% through structured **large-scale data transformation**, feature preparation, controlled evaluations, and documented analytical results for research stakeholders.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **AI model evaluation**, debugging, failure analysis, and collaborative research reviews.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Engineered cross-platform microservices and ETL integrations using AWS Glue, S3, REST APIs, and object-oriented patterns, improving pipeline efficiency 68%, response times 30%, and **distributed data reliability** across environments.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds, supporting **code reviews**, debugging, Git collaboration, and dependable releases across Windows, Linux, and macOS.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight MERN microservices for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **responsive React interfaces**, REST APIs, WebSockets, and robust validation across distributed locations.
- Developed TypeScript workflows for real-time records and third-party integrations, while Jenkins CI/CD, Docker staging, and **automated regression testing** reduced deployment time 50% across controlled production releases safely.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, C#, Bash, HTML5, CSS3
Application Development: React, React Native, Node.js, Spring Boot, Flask, FastAPI, REST APIs, Microservices, Responsive Design, Mobile Applications, OOP, Design Patterns, WebSockets
Cloud & Data: AWS, Azure, GCP, Docker, Kubernetes, PostgreSQL, SQL Server, MySQL, MongoDB, Data Modeling, ETL, ELT, Data Pipelines, FHIR, OAuth 2.0, Analytics
Engineering: Git, GitHub, CI/CD, Jenkins, GitHub Actions, DevOps, Unit Testing, Integration Testing, Regression Testing, PyTest, Mockito, JUnit, Code Review, Debugging, Agile, SDLC, Security, Technical Documentation, AI-Assisted Development

Projects

CareConnect Digital Front Door | *React Native, TypeScript, Java, Spring Boot, AWS, PostgreSQL*

- Designed a secure patient-access application for appointments, records, and virtual-care navigation, combining responsive mobile workflows, OAuth 2.0 authentication, **Java microservices**, relational data models, validation, and automated tests.

Clinical Event Pipeline | *Python, Kafka, Airflow, AWS, SQL, FHIR, Docker*

- Built a cloud-native pipeline that validates and transforms healthcare events into FHIR-aligned records, using **Python ETL workflows**, Kafka ingestion, Airflow orchestration, observability, retry handling, and containerized deployments.

Hospital Management Information System | *React, Node.js, PostgreSQL, FHIR, Sentry, Twilio*

- Built a healthcare management application with responsive React workflows, secure REST APIs, patient and appointment records, **FHIR interoperability**, Twilio notifications, PostgreSQL data models, and Sentry-based monitoring for reliable issue diagnosis.